
Problem 10.18. Prove that if $A \subseteq \{0,1\}^*$, a family of branching programs $(B_1, B_2, ...)$ exists wherein each B_n accepts exactly the strings in A of length n and is bounded in size by a constant times n.

Proof. A is a regular language. Therefore, a DFA D exists that accepts every string in A. Construct each B_n as follows.

1. States of D are query nodes in B_n, and transitions of D are outgoing edges in B_n.
2. Each input symbol w_i in string $w_1 w_2 \cdots w_n$ corresponds to variable x_i in B_n.
3. B_n includes all nodes and transitions of D required for accepting all strings of length n and loops are unrolled as many times as needed.
4. Start node in D is also the start node in B_n.
5. Each query node q in B_n is labeled x_{p+1}, where p is the length of path from start node to q.
6. Finally, accepts node of D are replaced by single output node labeled 1, and all reject paths end at output node labeled 0.

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